

The square of a sum and of a difference

Two formulae worth knowing by heart — and the middle term that everybody forgets.

LESSON 98–99

2 × 40 MIN

MATEMATIKA 7, §44

S8 2.3

LESSON CLOCK

15'

20'

25'

20'

5'

Where the formulae come from	15 min
The square picture	20 min
Using them both ways	25 min
Squaring numbers in the head	20 min
Homework	5 min

LEARNING OBJECTIVES

- Derive $(a + b)^2$ and $(a - b)^2$ by multiplying out.
- Apply both formulae in either direction.
- Explain the middle term with a square picture.
- Use the formulae to square numbers mentally.

TEXTBOOK

National	pp. 282–289
Cambridge	Stage 8, pp. 18–24
Workbook	Exercise 2.3

01

Explanation

Where the formulae come from

Nothing new is being claimed: both formulae are ordinary multiplications, done once and then remembered.

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

STEP	WORKING
write it out	$(a + b)(a + b)$
four products	$a^2 + ab + ba + b^2$
collect	$a^2 + 2ab + b^2$

$(A + B)^2$ IS NOT $A^2 + B^2$

Try $a = 3, b = 4$: the left side is 49, the right 25. The missing $2ab = 24$ is the whole difference, and it is missing from more answers than any other term in the course.

The square picture

A square of side $a + b$ splits into four pieces: two squares and two equal rectangles.

PIECE	DIMENSIONS	AREA
large square	$a \times a$	a^2
rectangle	$a \times b$	ab
rectangle	$b \times a$	ab
small square	$b \times b$	b^2

Total: $a^2 + 2ab + b^2$. The two rectangles are the middle term, and they are impossible to overlook once the picture has been drawn.

THE SIGN IS THE ONLY DIFFERENCE

In $(a - b)^2$ the middle term is subtracted, but b^2 is still added: a square is never negative. Both formulae end in $+ b^2$.

Using them both ways

EXPAND	ANSWER
$(x + 5)^2$	$x^2 + 10x + 25$
$(x - 7)^2$	$x^2 - 14x + 49$
$(3a + 2)^2$	$9a^2 + 12a + 4$
$(2x - 5y)^2$	$4x^2 - 20xy + 25y^2$

RECOGNISE A PERFECT SQUARE	AS A SQUARE
$x^2 + 10x + 25$	$(x + 5)^2$
$x^2 - 14x + 49$	$(x - 7)^2$
$9a^2 + 12a + 4$	$(3a + 2)^2$
$x^2 + 6x + 10$	not a perfect square

THE TEST FOR A PERFECT SQUARE

Take the square roots of the first and last terms, double their product, and compare with the middle term. For $x^2 + 6x + 10$ that gives $2 \cdot x \cdot \sqrt{10}$, which is not $6x$ — so it is not one.

Squaring numbers in the head

NUMBER	SPLIT	WORKING	ANSWER
51^2	$(50 + 1)^2$	$2500 + 100 + 1$	2601
49^2	$(50 - 1)^2$	$2500 - 100 + 1$	2401
102^2	$(100 + 2)^2$	$10000 + 400 + 4$	10404
98^2	$(100 - 2)^2$	$10000 - 400 + 4$	9604

SPLIT AT THE NEAREST ROUND NUMBER

Choose a to be the ten or hundred nearest to the number; then b is small, b^2 is tiny, and the whole calculation fits in the head.

02 Worked examples

EXAMPLE 1 Expand $(2x - 5y)^2$.

① $a = 2x, b = 5y$.

② $a^2 = 4x^2$

③ $2ab = 2 \cdot 2x \cdot 5y = 20xy$

Subtracted, because of the minus.

④ $b^2 = 25y^2$, so the answer is $4x^2 - 20xy + 25y^2$.

ANSWER

$$4x^2 - 20xy + 25y^2$$

EXAMPLE 2 Is $9a^2 + 12a + 4$ a perfect square?

① $\sqrt{9a^2} = 3a$ and $\sqrt{4} = 2$.

② $2 \cdot 3a \cdot 2 = 12a$

It matches the middle term.

③ So it is $(3a + 2)^2$.

ANSWER

$$(3a + 2)^2$$

EXAMPLE 3 Find 49^2 mentally.

① $49 = 50 - 1$

② $50^2 = 2500$

③ $2 \cdot 50 \cdot 1 = 100$, subtracted.

④ $2500 - 100 + 1 = 2401$

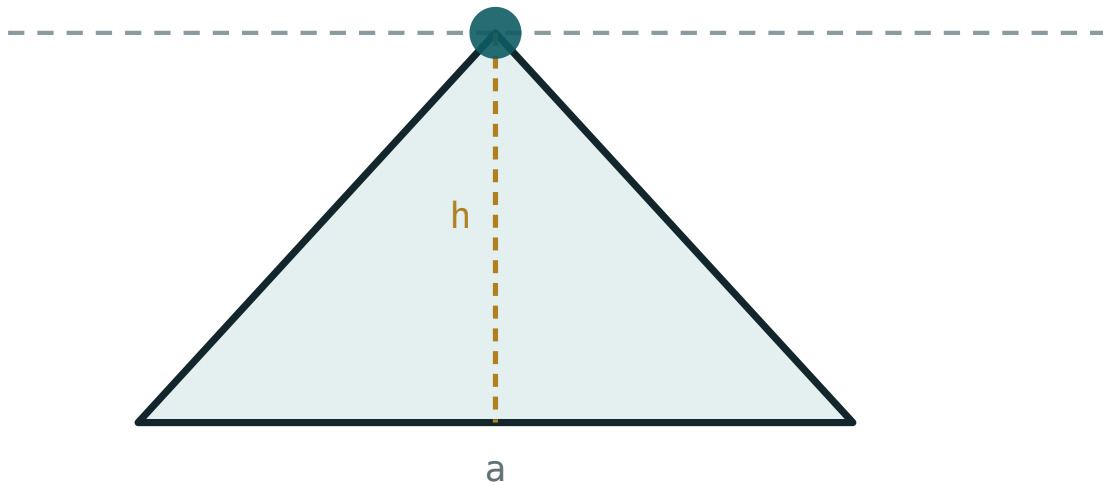
ANSWER

2401

03 Interactive model

Ask the class for $(a + b)^2$ before teaching anything; collect the wrong $a^2 + b^2$ answers on the board, then draw the square picture beside them.

The square cut into four pieces

base a **220**height h **120**area $\frac{1}{2}ah$ **13200**perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Square of a sum	Yig'indining kvadrati	Квадрат суммы
Square of a difference	Ayirmaning kvadrati	Квадрат разности
Abridged multiplication	Qisqa ko'paytirish	Сокращённое умножение
Middle term	O'rta had	Средний член
Double product	Ikkilangan ko'paytma	Удвоенное произведение
Identity	Ayniyat	Тождество
Perfect square	To'la kvadrat	Полный квадрат
Mental arithmetic	Og'zaki hisoblash	Устный счёт

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$(a + b)^2$ equals:

$$a^2 + b^2$$

$$a^2 + 2ab + b^2$$

$$a^2 - 2ab + b^2$$

$$2a + 2b$$

$(a - b)^2$ equals:

$$a^2 - b^2$$

$$a^2 - 2ab + b^2$$

$$a^2 + 2ab - b^2$$

$$a^2 - 2ab - b^2$$

$(x + 5)^2$ equals:

$$x^2 + 25$$

$$x^2 + 10x + 25$$

$$x^2 + 5x + 25$$

$$x^2 + 10x + 10$$

The middle term of $(3a + 2)^2$ is:

$6a$

$12a$

$5a$

$9a$

$x^2 + 6x + 10$ is:

a perfect square

not a perfect square

$(x + 3)^2$

$(x + 5)^2$

49^2 by the formula is:

2401

2409

2601

2500

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $(x + 3)^2$

ANSWER $x^2 + 6x + 9$

2. $(x - 4)^2$

ANSWER $x^2 - 8x + 16$

3. $(x + 5)^2$

ANSWER $x^2 + 10x + 25$

4. $(a + 1)^2$

ANSWER $a^2 + 2a + 1$

5. $(2x + 1)^2$

ANSWER $4x^2 + 4x + 1$

6. $(y - 6)^2$

ANSWER $y^2 - 12y + 36$

7. 51^2

ANSWER 2601

Medium · the standard question**7 PROBLEMS**

1. $(3a + 2)^2$

ANSWER $9a^2 + 12a + 4$

2. $(2x - 5y)^2$

ANSWER $4x^2 - 20xy + 25y^2$

3. $(4m - 3n)^2$

ANSWER $16m^2 - 24mn + 9n^2$

4. $x^2 + 10x + 25$ as a square

ANSWER $(x + 5)^2$

5. $x^2 - 14x + 49$ as a square

ANSWER $(x - 7)^2$

6. 98^2

ANSWER 9604

7. 102^2

ANSWER 10404

Hard · stretch and reason

7 PROBLEMS

1. $(a + b)^2 - (a - b)^2$

ANSWER $4ab$

2. $(a + b)^2 + (a - b)^2$

ANSWER $2a^2 + 2b^2$

3. If $a + b = 7$ and $ab = 12$, find $a^2 + b^2$

ANSWER 25

4. If $a - b = 3$ and $ab = 10$, find $a^2 + b^2$

ANSWER 29

5. $(x + \frac{1}{x})^2$

ANSWER $x^2 + 2 + \frac{1}{x^2}$

6. For which k is $x^2 + kx + 36$ a perfect square?

ANSWER $k = \pm 12$

7. $(2a - 3)^2 - (2a + 3)^2$

ANSWER $-24a$ **07 Homework****Homework — 5 tasks**

Write the middle term before the last one; it is the term that gets forgotten.

1. Expand $(x + 8)^2$, $(x - 9)^2$ and $(5a + 2)^2$.
2. Expand $(3x - 4y)^2$.
3. Write $x^2 + 12x + 36$ and $4a^2 - 20a + 25$ as squares.
4. Find 52^2 and 97^2 mentally, showing the split.

5. If $a + b = 9$ and $ab = 20$, find $a^2 + b^2$.